Module 3 — Periodic Table and Energy

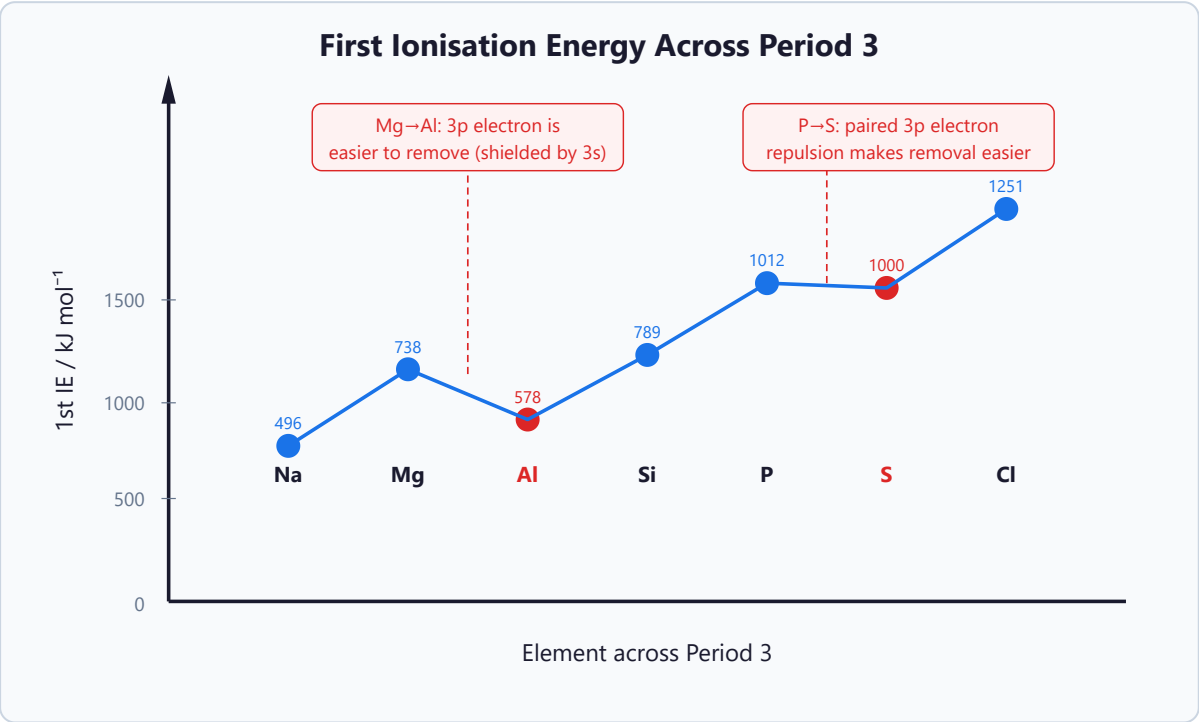
3.1 Periodicity

Periodicity: The repeating pattern of physical and chemical properties of elements across each period of the periodic table, caused by the repeating pattern of electron configurations. As each new period begins, a new principal energy level starts to fill, and the pattern of properties repeats.

Trends Across Period 3 (Na to Ar)

Property	Trend Across Period 3	Explanation
Atomic radius	Decreases Na → Ar	Increasing nuclear charge pulls electrons closer to the nucleus; shielding remains approximately constant (same number of inner shells), so the effective nuclear charge experienced by outer electrons increases.
First ionisation energy	Generally increases (with two anomalies)	Increasing nuclear charge with constant shielding means more energy is needed to remove an outer electron. Anomalies occur at Al (3p easier to remove than 3s) and S (paired 3p electron repulsion).
Melting point	Increases Na→Al, then Si very high, then drops sharply for P, S, Cl, Ar	Na, Mg, Al: metallic bonding strengthens (more delocalised electrons per atom and smaller ionic radius give stronger attraction). Si: giant covalent structure (very strong covalent bonds throughout). P_4, S_8, Cl_2, Ar: simple molecular structures (weak London forces between molecules; strength depends on M_r).

Electronegativity	Increases Na → Cl (Ar has no value as it does not form bonds)	Greater nuclear charge attracts bonding electrons more strongly. Same shielding across the period.



First Ionisation Energy — Definition and Anomalies

First ionisation energy: The energy required to remove one electron from each atom in one mole of gaseous atoms to form one mole of gaseous 1+ ions. This must be learned as a precise definition for exam questions.

$$X(g) \rightarrow X^{+}(g) + e^{-}$$

Factors affecting ionisation energy:

- **Nuclear charge:** More protons → stronger attraction → higher IE
- **Electron shielding:** More inner shells → outer electrons experience less nuclear charge → lower IE
- **Distance from nucleus:** Electrons further from nucleus are held less tightly → lower IE
- **Sub-shell and pairing:** Electrons in higher-energy sub-shells or paired electrons are easier to remove

Two anomalies occur across Period 3:

- **Mg → Al drop:** Al ($1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$) has its outer electron in a 3p orbital, which is higher in energy and further from the nucleus than the 3s orbital of Mg. The 3p electron is also partially shielded by the $3s^2$ electrons. Therefore less energy is needed to remove it.
- **P → S drop:** Sulfur ($3p^4$) has one pair of electrons in one of its 3p orbitals. The repulsion between the two paired electrons in the same orbital makes it easier to remove one compared to removing an unpaired electron from phosphorus ($3p^3$, which has a stable half-filled arrangement with all electrons unpaired).

Exam Tip: When explaining ionisation energy anomalies, you *must* mention: (1) the specific sub-shell involved (e.g. "3p" not just "outer shell"), (2) the specific factor making removal easier (shielding by $3s^2$ for Al, or electron pairing repulsion for S), and (3) compare to the previous element explicitly. Generic answers about "more protons" will not score marks for anomaly questions.

Successive Ionisation Energies

Successive ionisation energies always increase because each electron is removed from an increasingly positive ion (greater attraction). A large jump in IE occurs when an electron is removed from a shell closer to the nucleus (an inner shell), because these electrons experience much less shielding and are much closer to the nucleus. The pattern of jumps in successive IEs can be used to determine the group of an element.

Worked Example — Using Successive IEs:

An element has successive IEs (kJ mol^{-1}): 578, 1817, 2745, 11578, 14831...

There is a large jump between the 3rd and 4th IEs. This means removing the 4th electron requires much more energy — it comes from a new, inner shell with much less shielding. The element has 3 electrons in its outer shell, so it is in **Group 3**. (The element is aluminium.)

3.2 Group 2 — The Alkaline Earth Metals

Trends Down Group 2

Property	Trend	Explanation
Atomic radius	Increases	Additional electron shells increase the distance of outer electrons from the nucleus
First ionisation energy	Decreases	Outer electrons are further from the nucleus with more shielding from inner shells, despite increasing nuclear charge. The shielding and distance effects outweigh the extra protons.
Reactivity	Increases	Outer electrons are lost more easily (lower ionisation energy), so the metals are oxidised more readily
Melting point	Generally decreases (with anomalies due to crystal structure changes)	Metallic bond weakens as ionic radius increases and charge density decreases, so the attraction between cations and the delocalised electron sea is weaker

Reactions of Group 2 Elements

With water: Group 2 metals react with water to form a metal hydroxide and hydrogen gas. Reactivity increases down the group because ionisation energy decreases.

$$Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(aq) + H_2(g) \quad \text{(very slow with cold water; reacts vigorously with steam)}$$

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g) \quad \text{(steady reaction; fizzing observed)}$$

$$Ba(s) + 2H_2O(l) \rightarrow Ba(OH)_2(aq) + H_2(g) \quad \text{(vigorous; strongly exothermic)}$$

With dilute acid: All Group 2 metals react vigorously with acids to produce a salt and hydrogen.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Thermal Decomposition of Group 2 Carbonates and Nitrates

Group 2 carbonates decompose on heating to form the metal oxide and carbon dioxide:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

The temperature required for decomposition **increases** down the group. This is because larger cations have a lower charge density and therefore polarise the carbonate ion less. The less the carbonate ion is distorted, the harder it is to break the C–O bond and release CO_2. $MgCO_3$ decomposes at a relatively low temperature (~540 °C), while $BaCO_3$ requires a very high temperature (~1360 °C).

The same trend applies to Group 2 nitrates (decomposition temperature increases down the group), but nitrates decompose to give the metal oxide, nitrogen dioxide, and oxygen:

$$2Ca(NO_3)_2(s) \rightarrow 2CaO(s) + 4NO_2(g) + O_2(g)$$

Flame Tests for Group 2

Element	Flame Colour	Explanation
Calcium	Brick red / orange-red	The heat of the flame excites electrons to higher energy levels. When they fall back to their ground state, they emit light of a characteristic wavelength (colour). Each element has unique energy level spacings, producing a unique flame colour.
Strontium	Crimson red	
Barium	Green	

Solubility Trends

Hydroxides: Solubility **increases** down Group 2. $Mg(OH)_2$ is sparingly soluble (used as "milk of magnesia" antacid); $Ca(OH)_2$ is slightly soluble ("limewater"); $Ba(OH)_2$ is soluble.

Sulfates: Solubility **decreases** down Group 2. $MgSO_4$ is soluble (Epsom salts); $CaSO_4$ is sparingly soluble (gypsum); $BaSO_4$ is insoluble (used in "barium meals" for X-ray imaging of the gut because it is opaque to X-rays and is non-toxic due to its extreme insolubility).

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Exam Tip: Remember the solubility trends as opposites: hydroxides get MORE soluble going down, sulfates get LESS soluble going down. The insolubility of $BaSO_4$ is the basis of the test for sulfate ions.

3.3 Group 7 — The Halogens

Trends Down Group 7

Property	Trend	Explanation
Physical state at RTP	F_2 and Cl_2 are gases, Br_2 is a liquid, I_2 is a solid	More electrons means stronger London dispersion forces between molecules, requiring more energy to overcome
Colour	F_2 pale yellow, Cl_2 yellow-green, Br_2 red-brown, I_2 purple/grey	Colour deepens down the group
Electronegativity	Decreases	Bonding pair is further from the nucleus with more shielding in larger atoms
Reactivity (as oxidising agents)	Decreases	Larger atoms gain electrons less readily because the incoming electron is further from the nucleus and more shielded by inner electrons, resulting in a weaker attraction

Displacement Reactions

A more reactive halogen will displace (oxidise) a less reactive halide ion from solution. This demonstrates the trend in oxidising ability: $Cl_2 > Br_2 > I_2$.

	KCl(aq)	KBr(aq)	KI(aq)
Cl_2(aq)	No reaction	Cl_2 displaces Br^-; solution turns orange	Cl_2 displaces I^-; solution turns brown

Br_2(aq)	No reaction	No reaction	Br_2 displaces I^-; solution turns brown
I_2(aq)	No reaction	No reaction	No reaction

$$Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$$

Ionic equation: $Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$

Chlorine is the oxidising agent (it is reduced from 0 to −1); bromide is the reducing agent (it is oxidised from −1 to 0).

Disproportionation

Disproportionation: A reaction in which the same element is simultaneously oxidised and reduced — the element starts in one oxidation state and ends up in two different oxidation states.

Chlorine reacts with cold dilute sodium hydroxide in a disproportionation reaction:

$$Cl_2(aq) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$$

Chlorine starts at oxidation state **0**. In NaCl it is **−1** (reduced) and in NaClO it is **+1** (oxidised). This is disproportionation because the same element (Cl) is both oxidised and reduced.

Tests for Halide Ions

Add dilute nitric acid first (to remove interfering ions such as carbonate and sulfite that might form precipitates with silver), then add silver nitrate solution ($AgNO_3$):

Halide Ion	Precipitate Colour	Precipitate Formula	Solubility in Ammonia (NH_3)
Cl^-	White	AgCl	Dissolves in **dilute** NH_3
Br^-	Cream	AgBr	Dissolves in **concentrated** NH_3 only
I^-	Yellow	AgI	**Insoluble** in NH_3 (does not dissolve)

Exam Tip: Always state that **dilute nitric acid** is added first when describing the silver nitrate test. This prevents false positives from carbonate or sulfite ions (which would also form white precipitates with silver). If the precipitate colours are too similar to distinguish (white vs cream), use the ammonia solubility test to confirm.

3.4 Qualitative Analysis

Tests for Ions

Test For	Reagent / Method	Positive Result
Carbonate (CO_3^{2-})	Add dilute acid	Effervescence; gas turns limewater milky (CO_2 reacts with $Ca(OH)_2$ to form $CaCO_3$ precipitate)
Sulfate (SO_4^{2-})	Add dilute HCl then $BaCl_2$(aq)	White precipitate of $BaSO_4$ (insoluble)
Ammonium (NH_4^+)	Add NaOH(aq) and warm gently	Pungent gas produced; turns damp red litmus paper blue (NH_3)

3.5 Enthalpy Changes

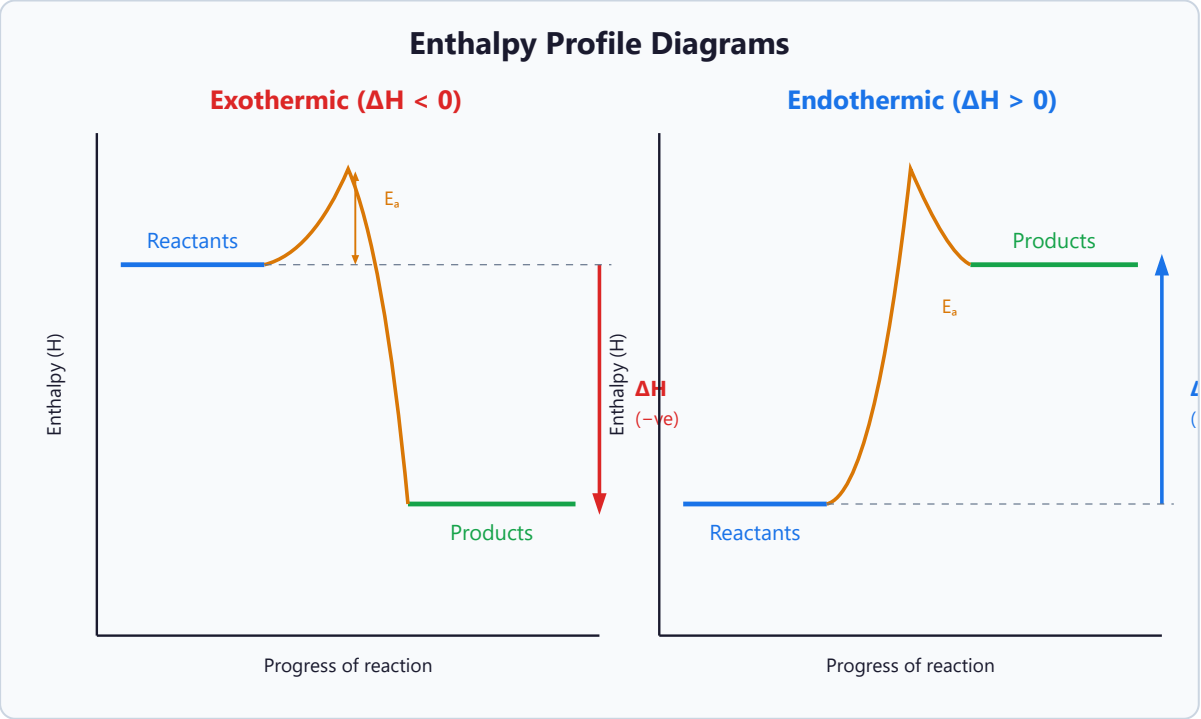
Enthalpy change (ΔH): The heat energy transferred at constant pressure during a chemical reaction. Measured in kJ mol^{-1}. **Exothermic** reactions have negative ΔH (heat released to surroundings; temperature of surroundings rises). **Endothermic** reactions have positive ΔH (heat absorbed from surroundings; temperature of surroundings falls).

Standard Enthalpy Definitions

Standard conditions: 100 kPa pressure, stated temperature (usually 298 K), substances in their standard states. The symbol ° indicates standard conditions.

Term	Symbol	Definition (learn precisely)
Standard enthalpy of combustion	$\Delta_c H°$	The enthalpy change when **one mole** of a substance undergoes **complete combustion** under standard conditions, with all reactants and products in standard states
Standard enthalpy of formation	$\Delta_f H°$	The enthalpy change when **one mole** of a compound is formed from its **elements in their standard states** under standard conditions
Standard enthalpy of neutralisation	$\Delta_{neut} H°$	The enthalpy change when an acid and a base react to form **one mole of water** under standard conditions
Standard enthalpy of atomisation	$\Delta_{at} H°$	The enthalpy change when **one mole of gaseous atoms** is formed from the element in its standard state

Exam Tip: These definitions *must* be learned precisely. Common errors: forgetting "one mole," forgetting "standard states," forgetting "gaseous atoms" for atomisation. The standard enthalpy of formation of any element in its standard state is zero by definition. $\Delta_c H$ is always negative (exothermic). $\Delta_{at} H$ is always positive (endothermic — bonds must be broken).



Calorimetry

$$q = mc\Delta T$$

Where q = heat energy transferred (J), m = mass of solution being heated (g), c = specific heat capacity (4.18 J g^{-1} K^{-1} for water), ΔT = temperature change (K or °C — the magnitude is the same). To find the enthalpy change per mole, divide q by the number of moles of the limiting reagent and express with the correct sign.

Worked Example — Calorimetry Calculation:

50.0 cm^3 of 1.00 mol dm^{-3} HCl is mixed with 50.0 cm^3 of 1.00 mol dm^{-3} NaOH. The temperature rises by 6.8 °C. Calculate $\Delta_{neut}H$.

Step 1: Total volume = 100.0 cm^3, so mass = 100.0 g (assuming density = 1.00 g cm^{-3})
Step 2: q = mcΔT = 100.0 × 4.18 × 6.8 = 2842 J = 2.84 kJ
Step 3: Moles of HCl = 1.00 × 50.0/1000 = 0.0500 mol (this equals moles of water formed)
Step 4: $\Delta_{neut}H$ = −2.84 / 0.0500 = **−56.8 kJ mol^{-1}**
(Negative because exothermic — temperature rose.)

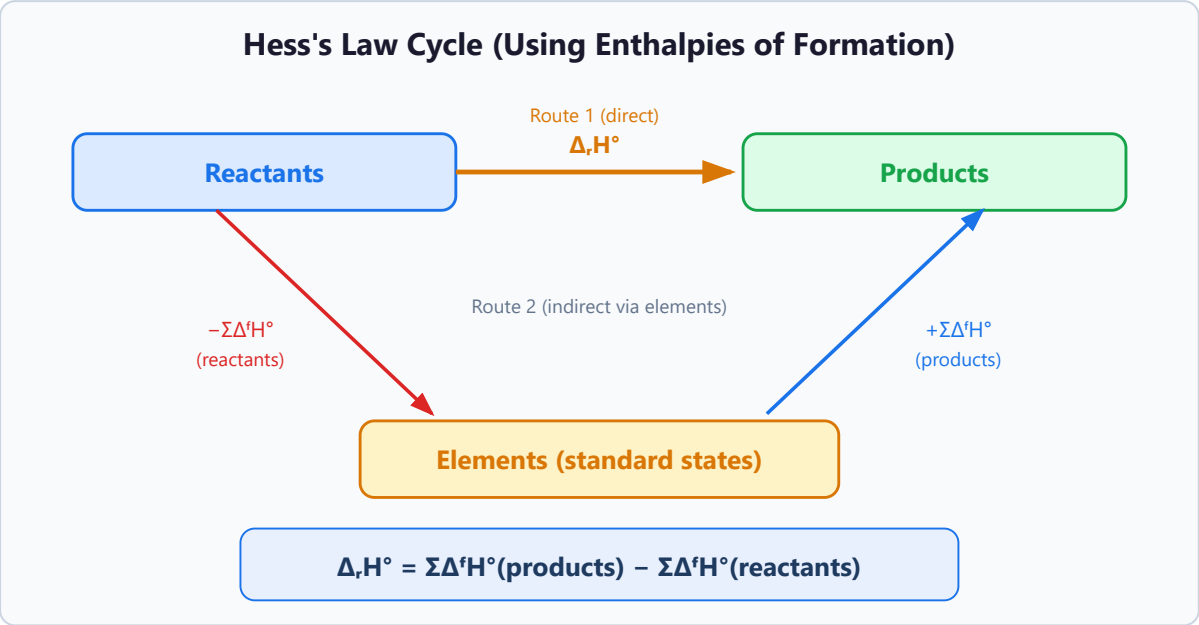
Exam Tip: In calorimetry calculations, always assume the density of the solution is 1.00 g cm^{-3} and the specific heat capacity is 4.18 J g^{-1} K^{-1} (same as water) unless told otherwise. Use the *total* volume of solution (both reactants combined) when calculating mass. Calorimetry typically gives inaccurate results because of heat loss to the surroundings — the experimental value is always less exothermic than the true value.

Hess's Law

Hess's Law: The total enthalpy change for a reaction is independent of the route taken, provided the initial and final conditions are the same. This is a consequence of the law of conservation of energy.

Using enthalpies of formation:

$$\Delta_r H^\circ = \Sigma\Delta_f H^\circ(\text{products}) - \Sigma\Delta_f H^\circ(\text{reactants})$$



Using enthalpies of combustion:

$$\Delta_r H^\circ = \Sigma\Delta_c H^\circ(\text{reactants}) - \Sigma\Delta_c H^\circ(\text{products})$$

Note the reversal: for combustion, it is reactants minus products (the opposite of the formation equation).

Worked Example — Hess's Law with Formation Enthalpies:

Calculate $\Delta_r H^\circ$ for: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

Given: $\Delta_f H^\circ(CH_4) = -74.8$ kJ mol^{-1}, $\Delta_f H^\circ(CO_2) = -393.5$ kJ mol^{-1}, $\Delta_f H^\circ(H_2O) = -285.8$ kJ mol^{-1}, $\Delta_f H^\circ(O_2) = 0$ (element)

$\Delta_r H^\circ = [(-393.5) + 2(-285.8)] - [(-74.8) + 2(0)]$

$= [-393.5 - 571.6] - [-74.8]$

$= -965.1 + 74.8 =$ **-890.3 kJ mol^{-1}**

Bond Enthalpies

Mean bond enthalpy: The average energy needed to break one mole of a specific type of bond in gaseous molecules, averaged across a range of different compounds. It is always positive (energy is required to break bonds).

$$\Delta_r H \approx \Sigma(\text{bond enthalpies of bonds broken}) - \Sigma(\text{bond enthalpies of bonds formed})$$

Bond enthalpy calculations give only **approximate** values because mean bond enthalpies are averages — the exact energy of a bond depends on the molecular environment (what other atoms are bonded nearby). The sign convention: breaking bonds is endothermic (+), forming bonds is exothermic (−).

Exam Tip: Remember: "energy in minus energy out" or "bonds broken minus bonds formed." If the total energy to break bonds exceeds the energy released in forming bonds, the reaction is endothermic. Bond enthalpy calculations only apply to reactions where all species are in the gaseous state. This is why Hess's law with formation or combustion enthalpies gives more accurate results.

3.6 Reaction Rates

Rate of reaction: The change in concentration of a reactant or product per unit time, typically measured in mol dm^{-3} s^{-1}.

Collision Theory

For a reaction to occur, particles must: (1) **collide** with each other, (2) with **sufficient energy** (greater than or equal to the activation energy, E_a), and (3) with the **correct orientation** so that reactive parts of the molecules come into contact. Factors that increase rate:

- **Increasing concentration** (or pressure for gases): More particles per unit volume means more frequent collisions per second, so more successful collisions per second.
- **Increasing temperature:** Particles have greater kinetic energy, so a significantly greater proportion of collisions exceed E_a. Collision frequency also increases slightly, but the main effect is the increase in the proportion of molecules with energy ≥ E_a.

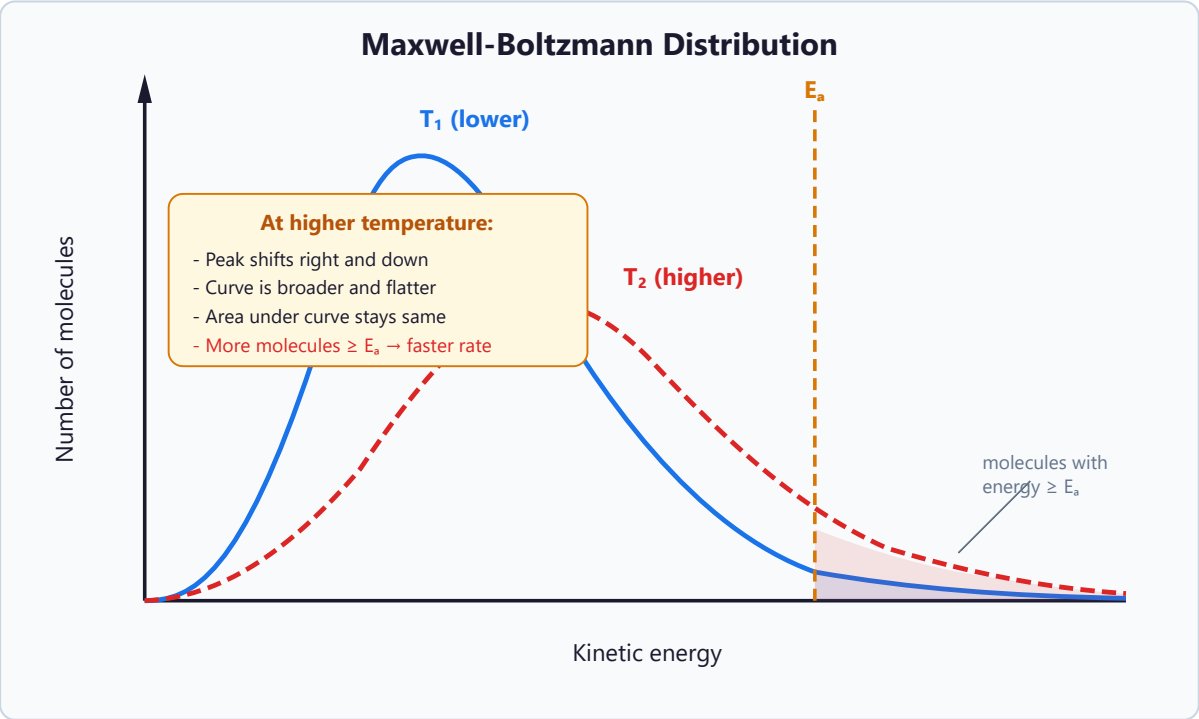
- **Increasing surface area:** More particles of a solid are exposed for collision (e.g. powder reacts faster than a lump of the same mass).
- **Using a catalyst:** Provides an alternative reaction pathway with a lower E_a. More particles have sufficient energy to react via this new pathway.

Maxwell-Boltzmann Distribution

The Maxwell-Boltzmann distribution shows the spread of molecular kinetic energies at a given temperature. Key features that *must* be shown correctly on a diagram:

- Starts at the origin (0,0) — no molecules have zero energy
- Rises steeply to a peak (the most probable energy)
- Tails off asymptotically at high energies (approaches but never touches the x-axis)
- The area under the curve represents the total number of molecules and is **constant**

At a **higher temperature**: the peak shifts right and down, the curve broadens and flattens, but the total area stays the same. Crucially, a much larger proportion of molecules have energy ≥ E_a, which is why even a small temperature increase causes a significant rate increase (roughly doubling the rate for every 10 K rise).



Exam Tip: When drawing Boltzmann distributions: (1) start at the origin, (2) the peak shifts right and down at higher temperature, (3) the area under both curves must be equal (same number of molecules), (4) the curve never touches the x-axis on the right. Label E_a on the x-axis and shade the area to the right of it. When explaining the effect of temperature, state that "a significantly greater proportion of molecules have energy greater than or equal to E_a, so there are more successful collisions per unit time."

Catalysts

Catalyst: A substance that increases the rate of a reaction by providing an **alternative reaction pathway with a lower activation energy**, without being consumed in the overall reaction. It is regenerated at the end.

On a Boltzmann distribution, a catalyst is shown by a new, lower E_a line to the left of the original. The distribution curve itself does *not* change. More molecules now have energy exceeding this lower activation energy, so the rate increases. On an enthalpy profile diagram, the catalyst produces a lower "hump" (activation energy) but the overall ΔH remains unchanged.

Catalysts can be **homogeneous** (same phase as reactants, e.g. H^+ in esterification) or **heterogeneous** (different phase, e.g. iron in the Haber process, vanadium(V) oxide in the Contact process). Heterogeneous catalysts work by adsorbing reactant molecules onto their surface, weakening bonds and holding molecules in the correct orientation.

3.7 Chemical Equilibrium

Dynamic equilibrium: A state reached in a **closed system** when the rate of the forward reaction equals the rate of the reverse reaction. The concentrations of reactants and products remain constant (but are not necessarily equal). Both reactions continue to occur — it is "dynamic" not static.

Conditions for dynamic equilibrium: (1) the system must be **closed** (no substances can enter or leave), (2) temperature must be **constant**, (3) both forward and reverse reactions are occurring at **equal rates**.

Le Chatelier's Principle

Le Chatelier's Principle: If a system at equilibrium is subjected to a change in conditions (concentration, pressure, or temperature), the position of equilibrium shifts in the direction that tends to counteract the change and partially restore the original conditions.

Change	Effect on Position of Equilibrium	Effect on K_c
Increase concentration of a reactant	Shifts right (towards products) to use up the added reactant	No change
Decrease concentration of a product	Shifts right (to replace removed product)	No change
Increase pressure (gaseous equilibrium)	Shifts to the side with **fewer moles of gas**	No change
Increase temperature (exothermic forward)	Shifts left (endothermic direction) to absorb excess heat	K_c **decreases**
Increase temperature (endothermic forward)	Shifts right (endothermic direction) to absorb excess heat	K_c **increases**
Add a catalyst	**No change** in position — reaches equilibrium faster	No change

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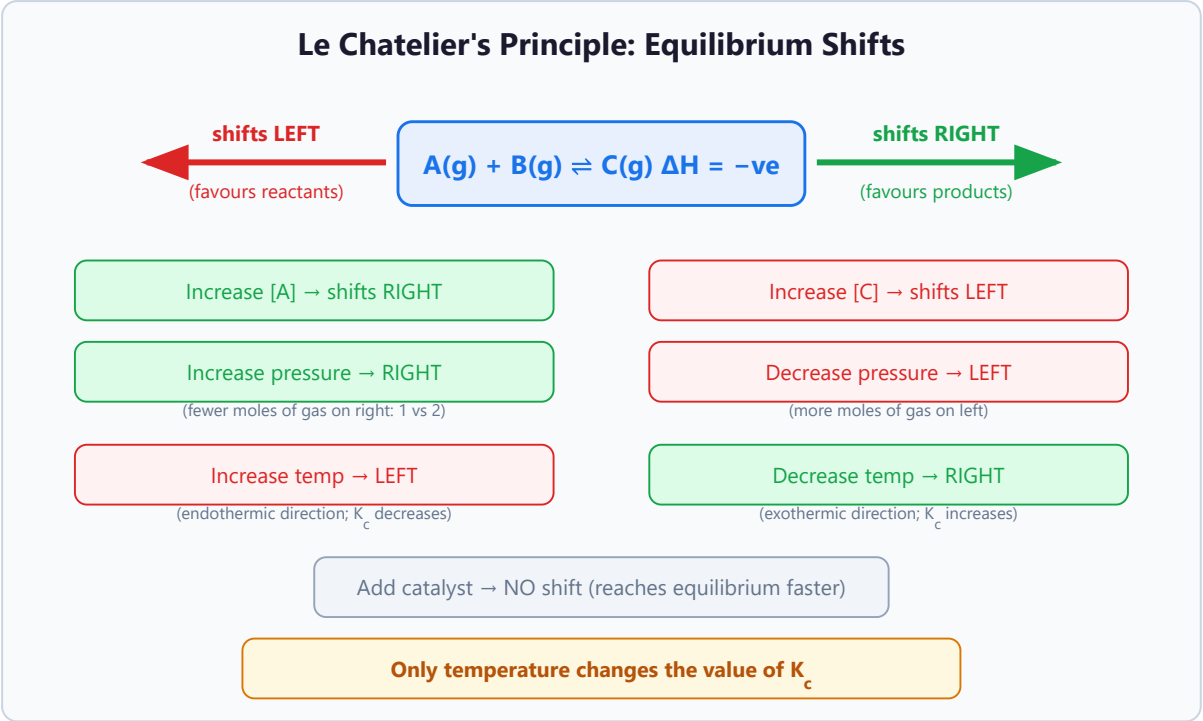

Equilibrium Constant K_c

For the reaction: aA + bB ⇌ cC + dD

$$K_c = [C]^c[D]^d / [A]^a[B]^b$$

K_c is **only affected by temperature**. A large K_c (>>1) means the equilibrium lies to the right (products favoured). A small K_c (<<1) means the equilibrium lies to the left (reactants favoured). Changes in concentration, pressure, or adding a catalyst do *not* change K_c.

Exam Tip: When calculating K_c, always use **equilibrium concentrations** (not initial concentrations). Set up an ICE table (Initial, Change, Equilibrium) to find equilibrium concentrations if they are not given directly. Always include the **units** of K_c, which depend on the overall change in moles. Only pure liquids and solids are excluded from the K_c expression.

Worked Example — Calculating K_c:

For $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, equilibrium concentrations at 700 K are: $[H_2]$ = 0.0200 mol dm^{-3}, $[I_2]$ = 0.0200 mol dm^{-3}, [HI] = 0.160 mol dm^{-3}.

$K_c = [HI]^2 / ([H_2][I_2]) = (0.160)^2 / (0.0200 \times 0.0200) = 0.0256 / 0.000400 =$ **64.0** (no units, as moles cancel)

K_c is large, confirming that at 700 K the equilibrium lies well to the right (products favoured).